Solar ESS Co. Ltd.
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PLATE FABRICATION · P2 COLUMN BASE PLATE TO MATCH THE PILE

FULL SIZE 1:1 when printed on A4 at 100 % (no "fit to page") · mm · SS400 · 36 pcs

0 — 100 mm: check this bar after printing

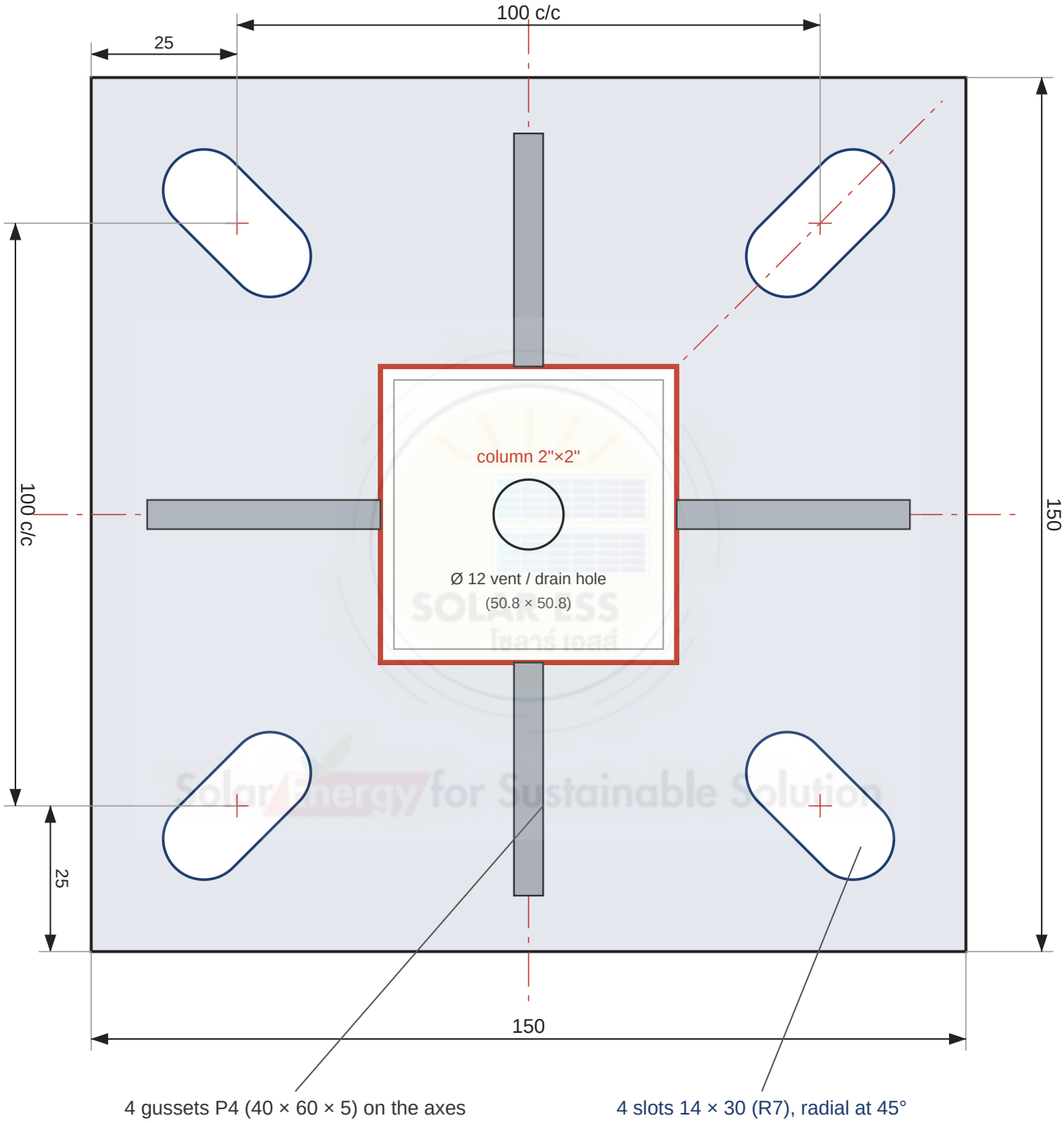


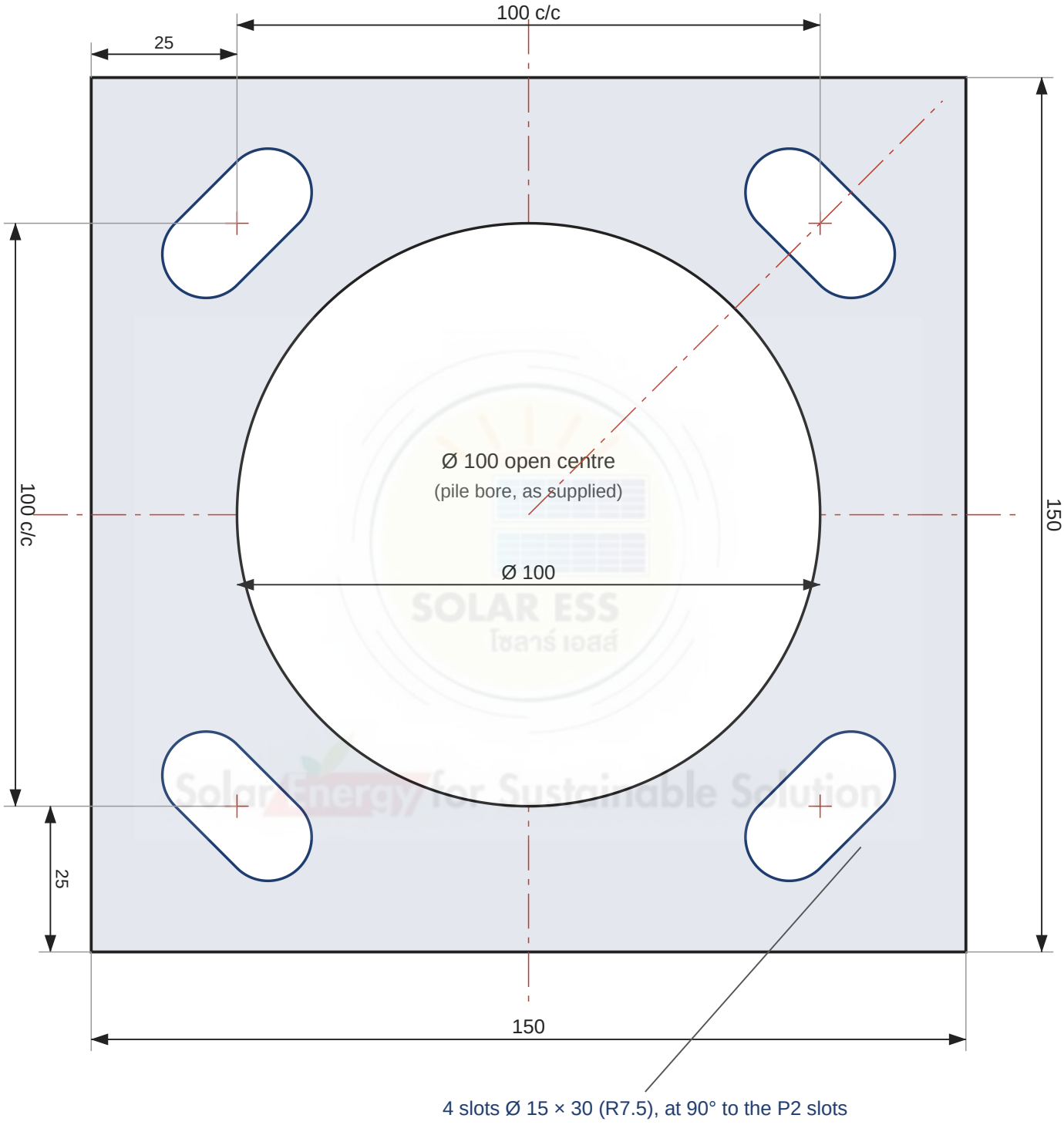
Plate 150 × 150 × 10, matches the pile head plate (150 / 100 c/c) · slots 14 × 30 pointing to the centre, i.e. at 90° to the pile-plate slots · Ø 12 vent hole at the centre (inside the column, needed for hot-dip galvanizing) · column 2"×2"×2.3 centred, fillet 3 all round · details on page 3

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P1 PILE HEAD PLATE · AS SUPPLIED ON THE SCREW PILE

FULL SIZE 1:1 at 100 % · mm · made by our supplier, welded on the Ø 100 × 1.5 m screw pile · do NOT fabricate · use it to check P2

0 100 mm: check this bar after printing

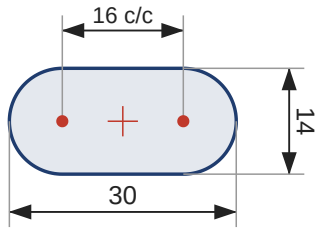


Supplier plate 150 × 150, Ø 100 open centre, 4 slots Ø 15 × 30 at 100 c/c, at 45° across the corners. P2 (page 1) has the same 150 / 100 c/c pattern with its slots turned 90° to these: laid together they give ± 7 mm adjustment both ways.

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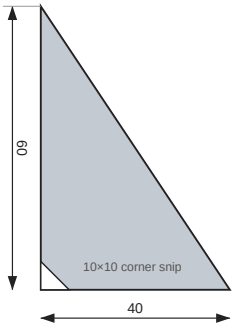
PLATE FABRICATION · DETAILS AND PARTS

Slot detail (1:1)



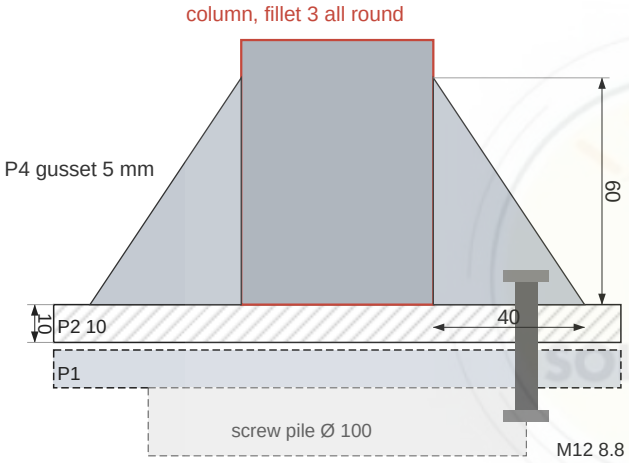
Drill 2 × Ø 14 at 16 c/c and cut out between, or plasma / laser.

P4 gusset (1:1)

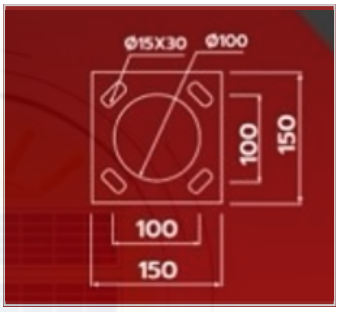


40 × 60 × 5 SS400, 4 per column.

Section through the base (1:2)



Pile head plate as supplied (reference)



Supplier plan: 150 × 150, Ø 100, slots Ø 15 × 30, 100 c/c



On site: open centre, 4 slots

Parts for 36 piles

Ref	Part	Size	Qty
P1	Pile head plate, Ø 100 open centre, 4 slots Ø 15 × 30 (on the pile, from our supplier)	150 × 150	36 *
P2	Column base plate, 4 slots 14 × 30 pointing to the centre + Ø 12 vent hole	150 × 150 × 10	36
P4	Gusset, 10 × 10 corner snip	40 × 60 × 5	144
P6	Plate washer, hole Ø 14	30 × 30 × 3	288
P8	Shim, slotted (1 / 2 / 3 mm)	150 × 150	as needed
—	Bolt M12 × 60 grade 8.8 HDG + nut + spring washer	M12	144

* P1 comes welded on the screw pile: not fabricated. Before cutting all 36 P2, lay the first P2 on a real pile head and check that M12 bolts pass at all 4 slot crossings. If the delivered pile plates have their slots parallel to the edges instead of at 45°, turn the P2 slots to 90° to them (vertical) before cutting the rest.

Fabrication notes

- Cut square to ± 1 mm; slot centres ± 0.5 mm. Mark from pages 1–2 printed at 100 % (check the 100 mm bar), or make one steel template and copy it.
- Deburr all edges and slots. Punch-mark each plate P1 or P2.
- Drill the Ø 12 vent hole at the P2 centre before welding the column (lets zinc in and out when galvanizing, drains water later).
- Weld column to P2 with a 3 mm fillet all round; gussets 3 mm both sides (E7018 or E6013).
- Hot-dip galvanize after welding, or zinc-rich primer + 2 coats cold galvanizing on every weld.
- Check: a P2 laid on a P1 turned 90° must let an M12 pass through all 4 slot crossings.
- Assembly: M12 × 60 gr 8.8 with a plate washer each side, tighten ≈ 70 N·m.